
Capstone Project: Inventory Management System

Objective:
Create a system to manage inventory levels, track stock movement, and identify products that need to be reordered or are overstocked.

Week 1 – Database Foundations: MySQL & MongoDB

Tools: MySQL, MongoDB

Capstone Tasks:
- Create MySQL tables: `products`, `warehouses`, `stock_movements`, `suppliers`
- Insert and update stock levels via CRUD operations
- Write a stored procedure to identify products below reorder level
- Use MongoDB to store audit logs or stock adjustment reasons in JSON
- Add indexes for quick search by product ID or warehouse

Deliverables:
- SQL script with schema, CRUD, stored procedure
- MongoDB script with audit logs and indexing

Week 2 – Stock Processing in Python

Tools: Python (Pandas, NumPy)

Capstone Tasks:
- Load stock movement logs (CSV or JSON)
- Clean and transform data (e.g., date formatting, quantity validation)
- Calculate current stock levels using `numpy`
- Use `pandas` to flag items below reorder threshold

Sample Code Snippet: ```python import pandas as pd import numpy as np

df = pd.readcsv("stockmovements.csv") df['quantity'] = df['quantity'].astype(int)

stocksummary = df.groupby('productid')['quantity'].sum() lowstock = stocksummary[stock_summary < 10]

print(low_stock) ```

Deliverables:
- Python script to compute current stock
- Summary report of low-stock items

Week 3 – PySpark for Warehouse-Level Insights

Tools: PySpark

Capstone Tasks:
- Load large stock movement data using PySpark
- Aggregate total stock per warehouse
- Identify warehouses with overstocked or understocked items

Deliverables:
- PySpark script with group and filter logic
- Output file with warehouse-level stock status

Week 4 – ETL in Azure Databricks

Tools: Azure Databricks

Capstone Tasks:
- Load cleaned stock and product data into Databricks
- Join product and warehouse info
- Create a master inventory view with reorder flag
- Save output in Delta or CSV format

Deliverables:
- Databricks notebook
- Final inventory file ready for dashboarding

Week 5 – Pipeline Automation with Azure DevOps

Tools: Azure DevOps

Capstone Tasks:
- Automate daily stock check via Azure DevOps pipeline
- Trigger Python script and export reorder list
- Output daily report of low-stock products

Deliverables:
- YAML pipeline to run Python job
- CSV log of products needing reorder

Final Outcome by Week 5:

- Inventory tracked via MySQL and MongoDB
- Cleaned and summarized data using Python and PySpark
- ETL processed in Azure Databricks
- Automated alerts for restocking via Azure DevOps
